

Let p be True. From (1) and defn. of conditional, q is True.
 From (5) and defn. of bi-conditional, $\neg(q \wedge r)$ is True. From (4) and defn. of conditional, $\neg s$ is True. From (3) and defn. of negation, $\neg(\neg(q \wedge r))$ is False. From the bi-conditional $\neg(\neg(q \wedge r)) \Leftrightarrow \neg p \wedge \neg s$ being True, by defn. of AND, $\neg p$ is False. By defn. of negation, p is True. By defn. of AND, $q \wedge r$ is True. By defn. of negation, $\neg(q \wedge r)$ is False which directly contradicts (2).

$\therefore p$ is False $\rightarrow$ Butler is surely lying

Let s be True. By defn. of conditional and (4), $\neg q$ is True. By defn. of negation, q is False. From (5) and defn. of biconditional, q is False. By defn. of AND, $q \wedge r$ is False. By defn. of negation, $\neg(q \wedge r)$ is True, which is consistent with (2).

By defn. of negation $\neg s$ is False. By defn. of AND, $(\neg p \wedge \neg s)$ is False, by defn. of negation, $\neg(\neg p \wedge \neg s)$ is True, which is consistent with (3).

$\therefore$ If s is True, then q is False and r is not determined

Let s be False. By defn. of conditional, $s \Rightarrow \neg q$ is True (matches with (4)). We have seen above that $(\neg p \wedge \neg s)$ is False. By defn. of negation, $\neg s$ is True. By defn. of AND, $\neg p$ is False. By defn. of negation, p is True. By defn. of negation, from (2), $\neg(\neg(q \wedge r))$ is False. From the biconditional $\neg(\neg(q \wedge r)) \Leftrightarrow (q \wedge r)$ being True, by defn. of AND, $q \wedge r$ is False. Since r is True, by defn. of AND, q is False.

$\therefore$ If s is False, then r is True and q is False

Finally p, q are always False. The truth values of r and s cannot be determined.

Final Answer: Both the Butler and the Cook are lying. About the Gardener and the Handyman, we are not sure